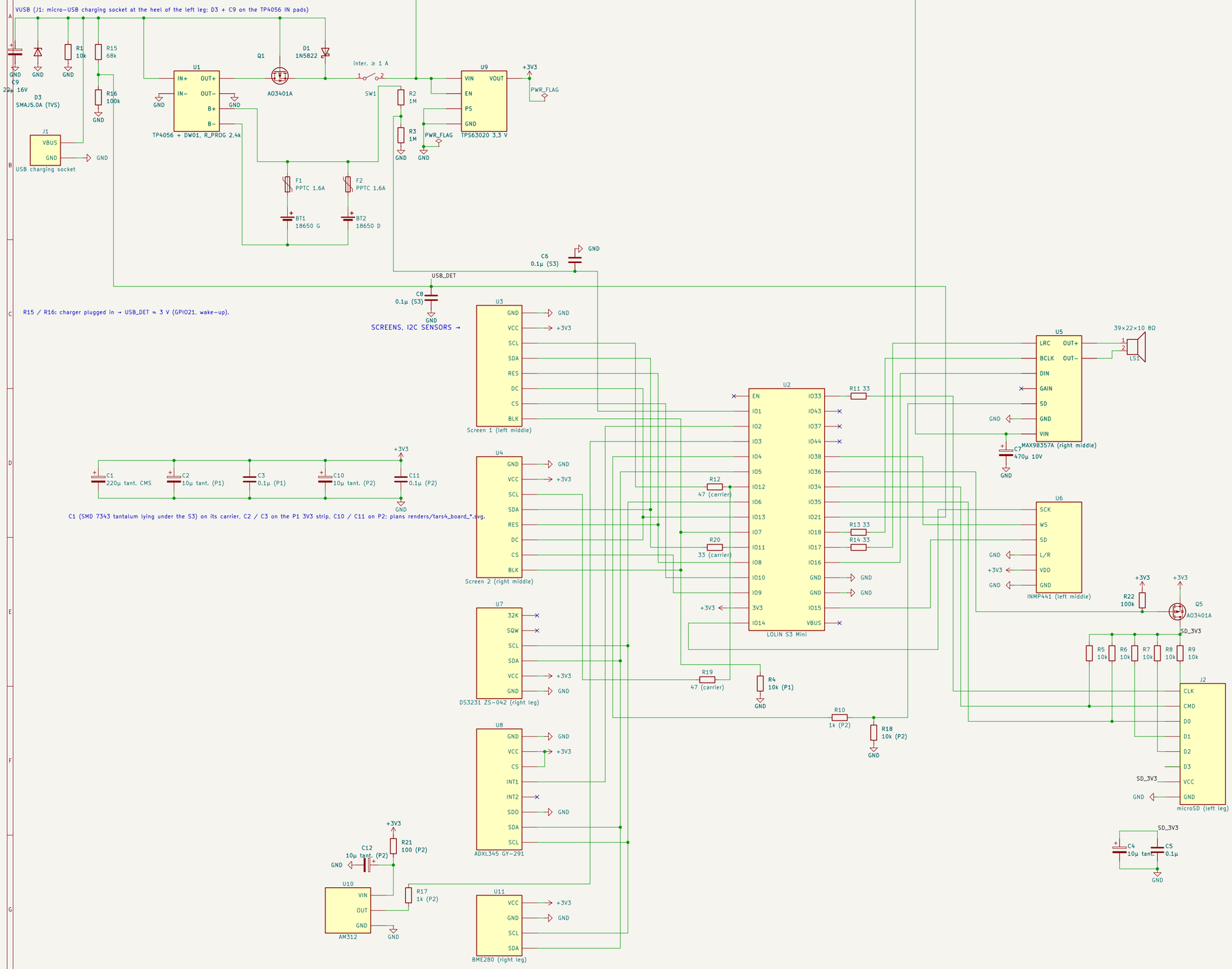
POWER — ground star on the power-path board (left leg)

BRAIN — LOLIN S3 Mini (left middle)

SOUND, MIC, microSD



U10: front hole at the top (right middle), R17 on P2; U7, U11 in the right leg (SQW not wired).

Q1 + D1 + R1: power path (mains / battery). R2 / R3 (1M, on B+: the real cell voltage, even on mains) on the power-path board; C6 (GPIO1) and C8 (GPIO21) at the S3 pins. Q5 + R22 on the power-path board: microSD cut in sleep (GPIO36 low = on).

U9: EN tied to VIN, PS/SYNC to GND (never drive it high). Its VSYS/GND pair leaves from the star; the logic GND to P1 leaves from its GND pad.

S3: V (VBUS) = the S3 USB-C only. EN = S3 reset (≠ BL_EN). 33 Ω series at the S3 (R20 on MOSI); SCK split on the carrier: R12 → screen 1, R19 → screen 2 (direct wire in the tube), 47 Ω (up to 100 if the image jitters).

BL_EN (GPIO7): backlight of both screens only (PWM possible). AMP_SD (GPIO4): R10 1k + R18 10k on P2 → amplifier SD (off at boot and in sleep). P1 / P2: stripboards under the screens. I2C chain: S3 → ADXL345 → middle tube → P2 → right tube → DS3231 + BME280 (right leg); sensor supply P1 → mic → ADXL345, P2 → right leg.

LEDs: remove every status LED fed by the battery (ZS-042, TPS63020, GY-291, microSD reader…); keep the TP4056 charge LEDs.

Silicone wire: power in 26 AWG (B+/B−, amplifier pair, 3V3 and GND to P1), signals and screen / sensor supply in 30 AWG. Tubes: 13 / 19 / 6 wires.

Connectors X1-X15 (keyed Dupont / JST) between each module and the harness, S3 plugged into a carrier: not drawn, see tars4_wiring.html. D3: unidirectional TVS (Zener symbol). A crossing without a dot is not a connection. Generated by tools/kicad_sch_gen.py (netlist checked at every generation).

LOLIN S3 Mini, TPS63020, 2 × 18650

Sheet: /
File: tars4.kicad_sch

Title: TARS4 — electronics

Size: A2	Date: 2026-10-21	**Rev: 3**
KiCad E.D.A. 10.0.4		Id: 1/1